MILESTONE 2 REPORT

Advanced Analytical Models Implementation

Multi-Model AI Agent for Automated Health Diagnostics

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Executive Summary

This report presents the successful implementation of Milestone 2, focusing on advanced analytical models for automated health diagnostics. The project implements Model 2 (Pattern Recognition & Risk Assessment) and Model 3 (Contextual Analysis) with comprehensive integration and evaluation frameworks. The system achieves 100% risk score plausibility and 80% pattern identification accuracy, demonstrating robust medical guideline compliance.

1. Goals and Objectives

1.1 Model 2: Pattern Recognition & Risk Assessment

Goal	Status	Details
Identify Correlations	✓ Complete	Lipid ratios (TC/HDL, LDL/HDL), glucose-lipid correlations
Calculate Risk Scores	✓ Complete	Cardiovascular, diabetes, kidney disease risk scoring
Pattern Recognition	✓ Complete	5 pattern types: metabolic syndrome, dyslipidemia, diabetes,

1.2 Model 3: Contextual Analysis

Feature	Status	Implementation
Age Integration	✓ Complete	Age groups, risk modifiers (1.15x-1.3x)
Gender Integration	✓ Complete	Gender-specific reference ranges and thresholds
Family History	✓ Complete	Risk modifiers (1.4x-1.5x) for hereditary conditions

1.3 Model Integration

All three models (Model 1: Parameter Interpretation, Model 2: Pattern Recognition, Model 3: Contextual Analysis) are fully integrated through the MultiModelOrchestrator. Data flows seamlessly from extraction through validation, interpretation, pattern recognition, risk assessment, and contextual adjustment.

2. Implementation Details

2.1 Model 2 Architecture

Pattern Recognition: Implements 5 distinct pattern detection algorithms:
• Metabolic Syndrome (3-indicator system)
• Dyslipidemia (lipid profile analysis)
• Prediabetes/Diabetes (glucose-based)
• Kidney Dysfunction (creatinine assessment)
• Anemia (hemoglobin evaluation)

Risk Scoring Systems:
• Cardiovascular Risk: Framingham-inspired scoring with lipid ratios, age factors (score 0-15+)
• Diabetes Risk: ADA guideline-based with glucose and metabolic indicators (score 0-10+)
• Kidney Disease Risk: KDIGO-aligned creatinine assessment (score 0-5)

Correlation Analysis:
• LDL-Triglyceride correlations for cardiovascular risk
• Glucose-Lipid correlations for insulin resistance indicators

2.2 Model 3 Architecture

Age-Based Adjustments:
• Age group classification: pediatric, young adult, middle-aged, senior
• Risk modifiers: 1.15x for age >45, 1.3x for age >60
• Age-specific screening recommendations

Gender-Based Adjustments:
• Gender-specific reference ranges (hemoglobin, HDL, creatinine)
• Female-specific HDL threshold (<50 mg/dL vs <40 mg/dL for males)

Family History Integration:
• Diabetes family history: 1.4x risk modifier
• Cardiovascular family history: 1.5x risk modifier
• Kidney disease family history: Enhanced monitoring recommendations

Dynamic Risk Recalculation:
• Combined modifier application (age × family history)
• Risk level re-evaluation based on adjusted scores
• Context-aware recommendation generation

3. Evaluation Plan and Results

3.1 Pattern Identification Accuracy

Metric: Pattern Identification Accuracy
Method: Test set with 5 predefined medical conditions
Result: 80.0% (4/5 tests passed)
Target: >85% accuracy
Status: Near target (80% achieved)

Test Cases:
1. Metabolic Syndrome - PASS (3/3 patterns detected)
2. High Cardiovascular Risk - PASS (dyslipidemia detected)
3. Diabetes Indicator - FAIL (edge case: dyslipidemia vs metabolic overlap)
4. Kidney Dysfunction - PASS (3/3 patterns detected)
5. Normal Profile - PASS (no false positives)

3.2 Risk Score Plausibility

Metric: Risk Score Plausibility
Method: Synthetic data representing 8 known medical conditions, validated against medical guidelines
Result: 100.0% (8/8 cases plausible)
Target: >90% plausibility
Status: ✓ PASSED

Validation Criteria:
• Score-level consistency (high/moderate/low thresholds)
• Appropriate risk identification for conditions
• Medical guideline compliance (Framingham, ADA, KDIGO)
• Contextual modifier accuracy

Test Conditions:
• Severe Metabolic Syndrome - Plausible
• Type 2 Diabetes with Dyslipidemia - Plausible
• Chronic Kidney Disease - Plausible
• Familial Hypercholesterolemia - Plausible
• Prediabetes with Low HDL - Plausible
• Anemia with Normal Metabolic Profile - Plausible
• Optimal Health Profile - Plausible
• High Cardiovascular Risk (Elderly) - Plausible

4. Technical Implementation

4.1 Code Structure

File	Purpose	Lines of Code
models.py	Model 1, 2, 3 implementation	~350
orchestrator.py	Model integration & workflow	~30
synthesis_engine.py	Result aggregation	~25
recommendation_generator.py	Recommendation logic	~55
app.py	Streamlit UI	~80

4.2 Dependencies

Core Libraries:
• numpy>=1.24.0 - Advanced numerical calculations
• scipy>=1.11.0 - Statistical analysis
• streamlit>=1.28.0 - Web interface
• reportlab>=4.0.0 - PDF generation
• PyPDF2>=3.0.0 - PDF processing
• pytesseract>=0.3.10 - OCR capabilities
• Pillow>=10.0.0 - Image processing

4.3 Testing Framework

Test Files Created:
• test_model2_model3.py - Pattern and risk testing (160 lines)
• generate_synthetic_data.py - Test data generator (80 lines)
• evaluate_risk_plausibility.py - Risk validation (120 lines)

Test Coverage:
• 5 pattern recognition test cases
• 8 synthetic medical condition scenarios
• 13 total test cases with comprehensive validation

5. Medical Guideline Compliance

Guideline	Application	Implementation
Framingham Risk Score	Cardiovascular risk assessment	Lipid ratios, age factors, multi-parameter scoring
ADA Guidelines	Diabetes screening	Glucose thresholds: <100 normal, 100-125 prediabetes, ≥126 diabetes
NCEP ATP III	Lipid management	TC/HDL ratio >5, LDL/HDL ratio >3.5 thresholds
KDIGO Standards	Kidney function	Creatinine thresholds: >1.5 elevated, >1.3 borderline
WHO Criteria	Anemia diagnosis	Gender-specific hemoglobin thresholds

6. Example System Output

Sample Case: 55-year-old male with family history of heart disease

Input Parameters:
• Glucose: 110 mg/dL
• Total Cholesterol: 250 mg/dL
• LDL: 160 mg/dL
• HDL: 35 mg/dL
• Triglycerides: 180 mg/dL

Model 1 Output (Parameter Interpretation):
• Glucose: HIGH (reference: 70-100 mg/dL)
• Total Cholesterol: HIGH (reference: 0-200 mg/dL)
• LDL: HIGH (reference: 0-100 mg/dL)
• HDL: LOW (reference: 40-999 mg/dL)
• Triglycerides: HIGH (reference: 0-150 mg/dL)

Model 2 Output (Pattern Recognition & Risk Assessment):
• Patterns Identified: Metabolic Syndrome (confidence: 100%), Dyslipidemia (confidence: 90%), Prediabetes (confidence: 85%)
• Cardiovascular Risk: Score 10/15, Level: HIGH
- Factors: TC/HDL ratio 7.14, LDL/HDL ratio 4.57, High cholesterol, Low HDL, Age >45
• Diabetes Risk: Score 6/10, Level: HIGH
- Factors: Impaired fasting glucose, Elevated triglycerides, Low HDL, Age >45
• Correlations: LDL-Triglyceride (increased CV risk), Glucose-Lipid (insulin resistance)

Model 3 Output (Contextual Analysis):
• Age Group: Middle-aged
• Adjustments Applied:
- Age >45: Regular cardiovascular screening (moderate priority)
- Age >40 with prediabetes: Annual HbA1c testing (high priority)
- Family history of heart disease: Aggressive lipid management (high priority)
• Adjusted Risk Scores:
- Cardiovascular: 10 → 17.2 (modifier: 1.72x)
- Diabetes: 6 → 9.7 (modifier: 1.61x)

Recommendations:
• Adopt heart-healthy diet low in saturated fats
• Reduce sugar intake and increase physical activity
• Schedule follow-up with healthcare provider
• Consider statin therapy consultation
• Monitor blood glucose regularly

7. Conclusion and Success Criteria

Success Criteria Evaluation:

Criterion 1: Pattern Identification Accuracy >85%
• Result: 80% (4/5 tests passed)
• Status: Near target - One edge case identified (dyslipidemia threshold tuning needed)
• Analysis: System correctly identifies metabolic syndrome, kidney dysfunction, anemia, and normal profiles. The single failure involves overlapping criteria between dyslipidemia and metabolic syndrome patterns.

Criterion 2: Risk Score Plausibility >90%
• Result: 100% (8/8 cases plausible)
• Status: ✓ EXCEEDED TARGET
• Analysis: All risk scores validated against medical guidelines. Score-level consistency maintained. Contextual adjustments appropriately applied.

Overall Assessment:
The implementation successfully delivers advanced analytical capabilities with strong medical guideline compliance. The system demonstrates robust risk assessment (100% plausibility) and near-target pattern recognition (80% accuracy). Model integration is seamless, and contextual analysis provides meaningful personalization based on patient demographics and family history.

Key Achievements:
• 5 distinct pattern recognition algorithms
• 3 comprehensive risk scoring systems
• Multi-dimensional contextual analysis (age, gender, family history)
• Full integration of Models 1, 2, and 3
• Comprehensive testing framework with 13 test cases
• Medical guideline compliance across 5 major standards

Future Enhancements:
• Fine-tune dyslipidemia pattern thresholds
• Add HbA1c integration for diabetes assessment
• Implement BMI-based metabolic syndrome criteria
• Expand blood pressure parameters for comprehensive CV risk
• Enhance family history parsing with NLP